

Measurement Matrix

Metrics:

- Force exerted when retrieving the snack (i.e. to open flap, drawer etc.) should not exceed 22N.
- An anti-theft system must be implemented such that no objects with a diameter larger than 12mm and length of 328mm can enter the storage area of the machine to dislodge a product (pass/fail)
- Hand flexion or extension during the process of retrieving the snack
- Time it takes for product retrieval

	Snack Chute	Snack Sack	Garage door	Wave n' take
Force (N)	0	15 ~ 20 N	5.60 N	0
Anti theft preservation (mm)	https://law.lis.virginia.gov/report/0MP7D/?utm_source=chatgpt.com https://www.pacodeandbulletin.gov/Display/pabull?d=reduce&file=%2Fsecure%2Fpabulletin%2Fdata%2Fvol32%2F32-8%2F297c.html&utm_source=chatgpt.com Gap between closed flap and border is 1.5mm Yes	Gap between closed flap and border is 1.5mm Yes	The maximum height any straight, rigid object can reach is 7mm away from the nearest snacks. Yes.	Gap between closed flap and border is 1.5mm Yes
Hand flexion / extension (°)	Average extension: 5.2 degrees Average flexion: 2.6 degrees	Average extension: 16 degrees Average flexion: 13.2 degrees	Average extension: 5.4 degrees Average flexion: 5.6 degrees	Extension: 7.2 Flexion: 9
Duration of retrieval process	Average duration: 2.515s	Average duration: 4.51 s	4.118s	4.16s